

Markscheme

Mathematics: analysis and approaches
Practice question bank

65. (a)

Three consecutive terms of an arithmetic sequence are $2x - 1$, $x + 5$, and $3x - 4$.

Find the value of x .

the middle term equals the average of its neighbours: $2(x + 5) = (2x - 1) + (3x - 4)$ (M1)

$$2x + 10 = 5x - 5 \quad \text{A1}$$

$$15 = 3x \quad (M1)$$

$$x = 5 \quad \text{A1}$$

[4 marks]

(b)

Hence find the common difference of the sequence.

the terms are 9, 10, 11 (M1)

$$d = 1 \quad \text{A1}$$

[2 marks]